① Aspect 14 (Most Central)

Make generalisations based on the results of applying mathematical procedures

Compared with qs1 and qs2, the defining feature of *ice-cream-shop-qs3* is that students are no longer required merely to compute a specific numerical answer. Instead, they must:

- Use previously obtained results
- Draw a **more general conclusion**, for example:
  - Which option is *generally* more advantageous
  - Under what conditions a particular choice is preferable

This represents a shift from "calculating results" to **using results to support broader judgments or generalisations**, making Aspect 14 the most prominent competency.

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② Aspect 12

Use a multi-step procedure leading to a mathematical solution, conclusion, or generalisation

To arrive at a valid generalisation, students typically need to:

- Revisit and integrate results from multiple calculation steps
- Organise information logically
- Base their conclusion on a coherent sequence of mathematical reasoning

Although the final output is a conclusion, it still rests on a **complete multi-step solution process**.

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③ Aspect 13

Use an understanding of the context to guide or expedite the solving process

Within the real-world **ice-cream shop context**, students must:

- Judge which differences are meaningful in practice
- Understand the implications of their conclusion for running the business
- Use contextual reasoning to justify *why* the conclusion makes sense

Here, contextual understanding plays a stronger role by **supporting and limiting the scope of the generalisation**.

④ Aspect 7

Manipulate numbers, graphical and statistical data and information

Even though qs3 focuses on conclusions, students must still:

- Correctly interpret and use numerical results obtained earlier
- Accurately manipulate these values when comparing, combining, or reasoning from them

Numerical manipulation therefore serves as a **tool in support of generalisation and judgment**.

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⑤ Aspect 8

Articulate a solution, showing and/or summarising intermediate mathematical results

This question typically requires students to:

- Clearly state their conclusion in words
- Briefly summarise the key calculations or results that support it

Clear communication of *how the conclusion was reached* is an important assessment criterion.

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⑥ Aspect 5

Understand and utilise constructs based on definitions, rules, and familiar algorithms

In forming generalisations, students implicitly rely on:

- Familiar quantitative relationships (e.g. “more than / less than”, totals, averages)
- Basic arithmetic rules as supporting tools

However, these are routine and not the main source of difficulty, so this aspect is less strongly represented.

① Aspect 14（最核心）

Make generalisations based on the results of applying mathematical procedures

📌 与 qs1、qs2 相比，*ice-cream-shop-qs3* 的关键特征在于：

- 不再只要求算出一个具体数值
- 而是基于前面得到的结果，**得出一个更一般性的结论**，例如：
  - 哪种方案"通常更有利"
  - 在什么条件下某种做法更合适

👉 这体现了从"算结果"上升到"用结果作判断或推广"，是本题最突出的能力，因此 Aspect 14 排在首位。

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② Aspect 12

Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation

📌 为了形成合理的概括，学生通常需要：

- 回顾或重新整合前面多步计算的结果
- 按逻辑顺序组织信息
- 在此基础上得出结论或一般性判断

👉 虽然最终输出是"结论"，但其基础仍然是一个**完整的多步骤求解过程。**

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③ Aspect 13

Use an understanding of the context to guide or expedite the solving process

📌 在冰淇淋店的现实情境中，学生需要：

- 判断哪些差异在现实中是重要的
- 理解结论在经营层面上的含义
- 结合情境解释"为什么这种结论是合理的"

👉 情境理解在 qs3 中的作用更加明显，用来**支撑和限制结论的适用范围。**

④ Aspect 7

Manipulate numbers, graphical and statistical data and information

📌 尽管 qs3 更偏向结论层面，学生仍然需要：

- 正确理解和调用之前得到的数值结果
- 在比较、综合或推断时准确操作这些数据

👉 数值操作在这里是**服务于概括与判断**的工具。

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⑤ Aspect 8

Articulate a solution, showing and/or summarising intermediate mathematical results

📌 本题通常要求学生：

- 用语言清楚表达结论
- 简要概括支持该结论的关键计算结果

👉 清晰表达"结论从何而来"是评价该题的重要维度。

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⑥ Aspect 5

Understand and utilise constructs based on definitions, rules and familiar algorithms

📌 学生在概括过程中隐含使用：

- 熟悉的数量关系（如"更多 / 更少""平均 / 总量"）
- 基本计算规则作为支撑

👉 但这些并非题目的主要挑战，因此相关性相对较低。